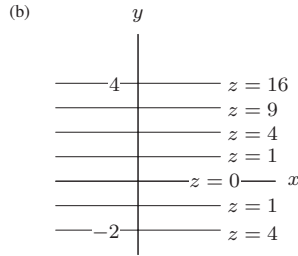
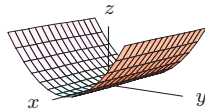
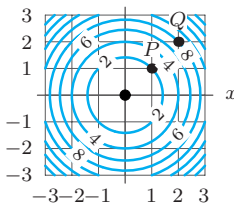


- 9 $\sin \theta \vec{i} + r \cos \theta \vec{j}$
 11 $\nabla z = \frac{1}{y} \cos\left(\frac{x}{y}\right) \vec{i} - \frac{x}{y^2} \cos\left(\frac{x}{y}\right) \vec{j}$
 13 $\left(\frac{-12\beta}{(2\alpha-3\beta)^2}\right) \vec{i} + \left(\frac{12\alpha}{(2\alpha-3\beta)^2}\right) \vec{j}$
 15 $60\vec{i} + 85\vec{j}$
 17 $10\pi\vec{i} + 4\pi\vec{j}$
 19 $(\pi/2)^{1/2} \vec{i}$
 21 $\frac{1}{100}(2\vec{i} - 6\vec{j})$
 23 $-46/5$
 25 $22/5$
 27 $84/5$
 29 $(2x + 3e^y)dx + 3xe^y dy$
 31 $(x+1)ye^x \vec{i} + xe^x \vec{j}$
 33 Negative
 35 Positive
 37 Positive
 39 $-\vec{j}$
 41 $\vec{j}$
 43 $-\vec{i} - \vec{j}$
 45 $\vec{i} - \vec{j}$
 47 (a) (3.96838, 5.09487)
 (b) 0.1052
 (c) $1/(3\sqrt{10})$
 49 4.4
 51 (a) $-16\vec{i} + 12\vec{j}$
 (b) $16\vec{i} - 12\vec{j}$
 (c) $12\vec{i} + 16\vec{j}$; answers may vary
 53 (a) -3.268
 (b) -4.919
 55 (a) Should be number
 (b) 11/5
 57 -0.5; better estimate is -1.35
 59 -1.1; better estimate is -1.8
 61 $4\vec{i} + 6\vec{j}$; $4(x-2) + 6(y-3) = 0$
 63 $-4\vec{i} + \vec{j}$; $-4(x-2) + (y-3) = 0$
 65 (a) Negative
 (b) Negative
 67 (a) Negative
 (b) Positive
 (c) Positive
 (d) Negative
 69 Yes
 71 Yes
 73 (a)



- (b)
 (c) $\vec{j}$
 75 (a) Circles centered at P
 (b) away from P
 (c) 1
 77 $(3\sqrt{5} - 2\sqrt{2})\vec{i} + (4\sqrt{2} - 3\sqrt{5})\vec{j}$
 79 $\frac{4\sqrt{2}}{6\vec{i} + 2\vec{j}}$
 81 $5/\sqrt{2}$
 83 (a) P, Q
 (b)

 (c) $\|\text{grad } f\|$
 $f_{\vec{u}} = \|\text{grad } f\| \cos \theta$
 89 $f_{\vec{u}}(0, 0)$ is scalar, not vector
 91 Closer contours give longer gradients
 93

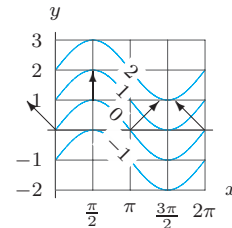


- 95 False
 97 False
 99 True
 101 False
 103 True
 105 True
 107 True
 109 True
 111 False

Section 14.5

- 1 $2x\vec{i}$
 3 $e^x e^y e^z (\vec{i} + \vec{j} + \vec{k})$
 5 $\frac{-2xyz^2}{(1+x^2)^2} \vec{i} + \frac{z^2}{1+x^2} \vec{j} + \frac{2yz}{1+x^2} \vec{k}$
 7 $(x\vec{i} + y\vec{j} + z\vec{k}) / \sqrt{x^2 + y^2 + z^2}$
 9 $y\vec{i} + x\vec{j} + e^z \cos(e^z) \vec{k}$

- 11 $e^p \vec{i} + (1/q)\vec{j} + 2re^{r^2} \vec{k}$
 13 $\vec{0}$
 15 $6\vec{i} + 4\vec{j} - 4\vec{k}$
 17 $-\pi\vec{i} - \pi\vec{k}$
 19 $9/\sqrt{3}$
 21 $-1/\sqrt{2}$
 23 $-\sqrt{77/2}$
 25 $-2\vec{i} - 2\vec{j} + 4\vec{k}$;
 $-2(x+1) - 2(y-1) + 4(z-2) = 0$
 27 $2\vec{j} - 4\vec{k}$; $2(y-1) - 4(z-2) = 0$
 29 $-2\vec{i} + \vec{k}$; $-2(x+1) + (z-2) = 0$
 31 (a) 0
 (b) $24/\sqrt{19}$
 33 (a) $(2x - yz)\vec{i} + (2y - xz)\vec{j} - xy\vec{k}$
 (b) $x + 4y + 6z = 20$
 35 $z = 12 - 4x - 2y$
 37 $6(x-1) + 3(y-2) + 2(z-1) = 0$
 39 $2x + 3y + 2z = 17$
 41 $z = 2x + y + 3$
 43 $x + 4y + 10z = 18$
 45 (a) is (III); (b) is (I); (c) is (IV); (d) is (II)
 47 $22/\sqrt{14}$
 49 (a) Spheres centered at the origin
 (b) $2x \sin(x^2 + y^2 + z^2) \vec{i} + 2y \sin(x^2 + y^2 + z^2) \vec{j} + 2z \sin(x^2 + y^2 + z^2) \vec{k}$
 (c) $0, 180^\circ$
 51 (a) $-\vec{j}$,
 $\frac{2}{3}\vec{i} - \frac{1}{3}\vec{j} + \frac{2}{3}\vec{k}$
 (b) $y = 0$; $2x - y + 2z = 3$
 (c) $x = y = 0$ and $z \neq 0$
 53 (a) $1.2x + 1.6y + 6z = 8$
 (b) Yes; $\pm(0.8, 0.6, 1)$
 55 (a)



- (b) Valley
 57 (a) $-4\vec{i} + 4\vec{j} + 2\vec{k}$
 (b) $24 \text{ mg/cm}^3/\text{sec}$
 59 $f_{\vec{u}_3} < f_{\vec{u}_2} < 0 < f_{\vec{u}_4} < f_{\vec{u}_1}$
 61 (a) Spheres centered at origin
 (b) $(-2x\vec{i} - 2y\vec{j} - 2z\vec{k})e^{-(x^2+y^2+z^2)}$
 (c) $-3\sqrt{2}e^{-1}$ degrees/sec
 63 (a) is (V); (b) is (IV); (c) is (V)
 71 $f_x(0, 0, 0)x + f_y(0, 0, 0)y + f_z(0, 0, 0)z = 0$
 73 $f(x, y, z) = 2x + 3y + 4z + 100$
 75 False
 77 False
 79 (a) $^\circ\text{C per meter}$
 (b) $^\circ\text{C per second}$
 (c) $^\circ\text{C per second}$

Section 14.6